Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.
- My student hour: Tuesdays 1.00pm - 2.00pm, Bragg building foyer.

Revision

- A function, f, from A to B is a subset of the Cartesian product $A \times B$ such that for each element $a \in A$, there is **exactly one** ordered pair in f with a as its first element.
- The notation $f : A \to B$ means f is a function with A as its **domain** and B as its **codomain**.
- If $f : A \to B$ and $g : B \to C$ are functions, then $g \circ f$ is the function which takes each element $a \in A$ to $g(f(a))$.
- A function $f : A \to B$ is **injective** (**one-to-one**) if the only way to satisfy $f(a_1) = f(a_2)$ is $a_1 = a_2$.
- A function $f : A \to B$ is **surjective** (**onto**) if for every element $b \in B$ there is an element $a \in A$ such that $f(a) = b$.
- A function $f : A \to B$ is **bijective** if it is injective and surjective.

Properties of functions

Property	**Definition**	**Characterisation via arrows**	**Characterisation via ordered pairs**
$f : A \to B$ is a function.	Each input is mapped to exactly one output.	Each element of A has exactly one arrow coming out out of it.	For every $a \in A$ there is exactly one ordered pair in f with a as the first element.
$f : A \to B$ is injective.	$f(a_1) = f(a_2)$ implies $a_1 = a_2$.	Different arrows go to different locations.	If (a_1, b) and (a_2, b) both belong to f then $a_1 = a_2$.
$f : A \to B$ is surjective.	The image of f is equal to B.	Every element of B has an arrow pointing to it.	For every $b \in B$ there exists an $a \in A$ such that (a, b) belongs to f.

Revision

- Let A be an arbitrary set. The **identity function** on A maps every element of A to itself. It is denoted by id_A.
- An **inverse** of the function $f : A \to B$ is a function $f^{-1} : B \to A$ such that

$$f^{-1} \circ f = \mathrm{id}_A \qquad \text{and} \qquad f \circ f^{-1} = \mathrm{id}_B$$

- Not every function has an inverse.
- If f has an inverse, then it is obtained by "flipping" every ordered pair in f.

$$f^{-1} = \{(b, a) \mid (a, b) \in f\}$$

When does a function have an inverse?

Theorem 6.9. Let $f : A \to B$ be a function. Then f has an inverse if and only if it is a bijection.

Relations

Relations are similar to functions, but we relax the condition that each input has to correspond to exactly one output.

Definition 7.1. Let A and B be sets. A **relation** from A to B is a subset of the Cartesian product $A \times B$.

Say that $R \subseteq A \times B$ is a relation. We write aRb to mean that (a, b) is in R, and $a \not R\ b$ to mean (a, b) is not in R.

Relations on a set

Very frequently, it is useful to think about relations from a set to itself. This is the case that $A = B$. A subset of $A \times A$ is called a **relation on** A.

Directed graphs

Relations on a set A can sometimes be illustrated with a **directed graph**. The elements of A are represented by dots. Each ordered pair in the relation is represented by an arrow from a dot to a dot. An ordered pair of the form (a, a) is represented by a **self-directed arrow** (a **loop**).

Properties of relations

Definition 7.2. Let A be a set and let $R \subseteq A \times A$ be a relation on A. If (a, a) is contained in R for every $a \in A$, then we say R is **reflexive**.

A directed graph represents a reflexive relation if every element has a self-directed arrow.

Properties of relations

Definition 7.3. Let A be a set and let $R \subseteq A \times A$ be a relation on A. Assume that whenever (a, b) is contained in R, then (b, a) is contained in R. Then we say that R is **symmetric**.

A directed graph represents a symmetric relation if every arrow is two-way. (Self-directed arrows are automatically two-way.)

Properties of relations

Definition 7.4. Let A be a set and let $R \subseteq A \times A$ be a relation on A. Assume that whenever (a, b) and (b, a) are both contained in R, then $a = b$. Then we say that R is **anti-symmetric**.

A directed graph represents an anti-symmetric relation if there are no two-way arrows (except for loops).

Properties of relations

Note that “anti-symmetric” is not the same as “not symmetric”!

Properties of relations

Definition 7.5. Let A be a set and let $R \subseteq A \times A$ be a relation on A. Assume that whenever (a, b) and (b, c) are both contained in R, then (a, c) is contained in R. Then we say that R is **transitive**.

Transitivity means whenever we can get from a to c in two steps, we can get there in one step.

Properties of relations

Property	**Definition**	**Characterisation via arrows**
Reflexive	$(a, a) \in R$ for all $a \in A$	Every element has a loop
Symmetric	$(a, b) \in R$ implies $(b, a) \in R$	Every arrow is two-way
Anti-symmetric	$(a, b) \in R$ and $(b, a) \in R$ implies $a = b$	The only two-way arrows are loops
Transitive	$(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$	If we can get from a to c in two steps, we can get there in one step

Equivalence relations

Definition 7.6. Let A be a set and let $R \subseteq A \times A$ be a relation on A. If R is reflexive, symmetric, and transitive, then R is an **equivalence relation**.

Example. Let n be a positive integer. Let $R \subseteq \mathbb{Z} \times \mathbb{Z}$ be the relation on the integers such that (a, b) is in R if and only if $a - b$ is divisible by n.

Equivalence classes

Definition 7.8. Let R be an equivalence relation on the set A. If a is an element of A then $\{x \in A \mid aRx\}$ is the set of all elements x such that (a, x) is in R. This set is called an **equivalence class**. We denote it by $[a]$.

Partitions

Definition 7.7. Let A be a set and let $\mathcal{P}$ be a set of non-empty subsets of A. If every element of A is in exactly one set that is in $\mathcal{P}$ then we say that $\mathcal{P}$ is a **partition** of A.

Equivalence classes

Proposition 7.9. Let R be an equivalence relation on the set A. Let a and b be elements of A. Then aRb if and only if $[a] = [b]$.

Theorem 7.10. Let R be an equivalence relation on the set A. Then $\{[a] : a \in A\}$ is a partition of A.

Partial orders

Definition 7.11. Let A be a set and let $R \subseteq A \times A$ be a relation on A. If R is reflexive, anti-symmetric, and transitive, then R is a **partial order**.

Partial orders

Proposition 7.12. Let A be a set and let $R \subseteq \mathcal{P}(\mathcal{A}) \times \mathcal{P}(\mathcal{A})$ be the relation on subsets of A such that (X, Y) is in R if and only if X is a subset of Y. Then R is a partial order.

Hasse diagrams

Partial orders can be represented by diagrams called **Hasse diagrams**.

A Hasse diagram can be obtained from the directed graph diagram by the following steps.

- Since the relation is reflexive, we know that every element has a loop. Therefore we can omit loops.
- Since the relation is anti-symmetric it cannot contain any **directed cycles** (Proposition 7.13). So we can draw the directed graph with every arrow directed upwards on the page. Therefore we can omit arrowheads.
- Since the relation is transitive, we can omit any arrows implied by transitivity.

Hasse diagrams